Problem Chosen	2025 HiMCM Summary Sheet	Team Control Number
B		16739

Quantitative Analysis of Urban Location Characteristics, Environmental Impacts, and Policy Strategy Effects on Mega Sporting Events

Sport mega-events create substantial social and economic value but also impose significant environmental costs. Therefore, it is of great importance to understand the environmental impacts of these events and to explore how changes in their planning may alter these impacts.

Firstly, taking Super Bowl LIX in New Orleans as the baseline, 17 quantitative indicators covering both environmental impacts and city location characteristics are established. Reasonable value ranges and physical interpretations specified based on published literature and statistical data. In addition, 8 sub-criteria are introduced to capture sustainability factors relevant to multi-sport competition.

Secondly, the relationship between city location characteristics and environmental impacts is quantitatively characterized. City features are first reduced by **principal component analysis (PCA) based on Pearson correlations**, and a **multiple linear regression model** is then constructed with the principal component scores as explanatory variables, yielding a **PCA-based urban environmental impact model**. Three latent dimensions—infrastructure level (PC1), climate–geographical conditions (PC2), and population density (PC3)—jointly explain about 85% of the total variance. Grid carbon intensity, renewable energy share, public transit coverage, and the share of green or low-carbon travel are identified as the main structural drivers of environmental performance.

Thirdly, an **environmental impact evaluation model** is developed within this multi-indicator framework. **ARIMA models** are applied to forecast future city characteristics for U.S. cities that have already hosted the Super Bowl and to project 2029 features for three NFL cities that have not yet hosted—Charlotte, Buffalo, and Kansas City. The forecasted features are then input into the principal component regression model to obtain future environmental indicators, which are further aggregated into composite scores using the **entropy-weight method**. The results indicate that **Santa Clara** exhibits the best environmental performance among historical hosts, while **Kansas City** attains the highest score among the three potential new hosts.

Fourthly, the model is extended to comprehensive mega-events such as the **Summer Olympic Games**. Building on the single-event evaluation framework, additional indicators reflecting the sustainability of multi-venue, multi-sport events are introduced, and **a composite venue evaluation model based on principal component regression is developed**. The results show that for single events, environmental pressure is concentrated mainly in short-term peaks in water use and spectator transport emissions, whereas for multi-venue, multi-sport events, **long-term effects** from Olympic Village energy consumption, venue network layout, and post-event venue reuse play a more dominant role. After applying the entropy-weight method to derive composite environmental scores, **Los Angeles** emerges as the best-performing host under the comprehensive event scenario.

Finally, a sensitivity analysis is conducted to quantify the influence of key policy variables on environmental impacts. The infrastructure investment, energy procurement, and transport-mode shifts are abstracted as **three policy control variables and mapped**, via **interpretable transfer functions**, to changes in city characteristics and environmental indicators. The resulting effects are then evaluated using an environmental composite score within the broader assessment model, confirming the robustness of the approach.

Keywords: Mega sporting events; Environmental impact; PCA regression; Entropy-weight method

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1 Introduction

1.1 Problem Background

As sport mega-events such as the Super Bowl and the Olympic Games generate substantial social and economic benefits, they also produce significant environmental impacts, especially in terms of energy use and carbon emissions. With the global push toward carbon neutrality and sustainable development, the tension between sport mega-events and environmental protection has become increasingly pronounced.

In recent years, organizing committees and third-party institutions have attempted to quantify the environmental impacts of these events through carbon-footprint accounting and environmental rating systems. However, the resulting scores are still used mainly for post-event summaries and image promotion. They are rarely incorporated into key decision stages such as host-city selection and infrastructure planning, so environmental impacts continue to receive insufficient weight in practice.

Against this background, it is necessary to construct a mathematical model that explicitly aims to minimize environmental impacts and enables meaningful comparison across different potential host cities. Taking the Super Bowl as a case study, this paper analyzes the structure of its environmental footprint and develops an evaluation model for environmentally sustainable venue selection, providing quantitative and operational decision support for organizing committees.

1.2 Restatement of the Problem

(1) Analyze the environmental impacts of the Super Bowl in New Orleans, focusing on energy, emissions, water use, and waste, and show how changing the host city would affect these factors.

(2) Develop a model that evaluates Super Bowl host cities based on city characteristics (electricity mix, climate, transport) and event parameters (attendance, energy demand, travel).

(3) Apply the model to historical Super Bowl hosts and select the most sustainable city for 2029, while evaluating three new NFL cities.

(4) Extend the model to other mega-events like the Olympics, considering factors like event duration, venue networks, and international travel.

(5) Summarize findings in a 1–2-page letter to the NFL, recommending the most sustainable 2029 host city and explaining how this model aligns with their sustainability goals.

1.3 Our work

In this paper, we develop a data-driven framework for evaluating the environmental performance of Super Bowl host cities. We construct a two-layer indicator system that separates event-level outcomes (energy use, emissions, water, and waste) from city-level drivers (climate, transport, and energy mix), using New Orleans as a reference case. We apply principal component analysis (PCA) and principal component regression (PCR) to link city characteristics to environmental indicators. We forecast city data to 2029 using ARIMA, aggregate the indicators into a sustainability score via entropy weighting, and apply the model to historical Super Bowl cities and new NFL cities, ranking them on environmental performance. Finally, we extend the framework to the Summer Olympics and conduct sensitivity analysis on infrastructure, energy, and transport before presenting a recommendation to the NFL.

2 Assumptions and Justifications

To simplify the problem, we make the following basic assumptions, each of which is properly justified.

- **Assumption 1: The historical climatic conditions of each candidate city during the Super Bowl period are relatively stable.**

Justification: This assumption presupposes that while there are regional differences in temperature and precipitation levels among candidate cities during the event period (typically February), the fluctuations in their historical data fall within a reasonable range and can serve as stability parameters in the model.

- **Assumption 2: The urban population flow patterns remain stable.**

Justification: Based on the regularities of historical data on urban population migration, this assumption holds that each candidate city maintains relatively stable population mobility characteristics on an annual time scale.

- **Assumption 3: Reasonable extrapolation can be conducted based on historical data trends**.

Justification: Due to practical constraints on data availability, the quantitative analysis in this paper is mainly constructed based on historical observation data up to 2024. We assume that the basic correlation patterns between urban characteristics and environmental impacts revealed by current data, the trajectory characteristics of infrastructure development, and the evolution trends of energy structure will remain consistent in the short term.

- **Assumption 4: Candidate cities can maintain system stability within the prediction period.**

Justification: We assume that within the model's prediction time horizon (up to 2029), none of the candidate cities will experience major sudden events that could significantly alter their infrastructure characteristics, environmental carrying capacity, or population distribution patterns.

3 Indicator Establishment

Table 1 Environmental Impact Indicators Based on the Super Bowl LIV in New Orleans

Category	Quantitative Indicator	Unit
Energy consumption	Power consumption of the venue	MWh
	Hotel electricity consumption intensity	kWh
Use of water	Water for venue maintenance	gallon (gal)
	per capita water consumption	gallon (gal)
Waste generation	Total amount of garbage on the competition day	ton(T)
	Recyclability/Biodegradability Rate	%
Greenhouse gas emission	Total carbon emissions of the audience	t CO_2e
	Total transportation carbon emissions	t CO_2e
	Materials and promotional carbon emissions	t CO_2e

Energy consumption is at the core of the event's environmental cost. Operating the Super Bowl—keeping venues and hotels supplied with power and heating—can consume over 110 million kWh of

energy [1]. For Super Bowl LIV in New Orleans, an average February temperature of only 13°C still required continuous heating to maintain comfort for 65,719 spectators and staff, leading to a substantial energy load. If this demand is met mainly by fossil-fuel-based grids, it directly results in thousands of tons of CO_2 emissions, making energy use one of the primary sources of the event's environmental impact.

Water use is mainly reflected in venue-related consumption and per-capita usage, with toilet flushing being the most concentrated source. During the halftime break, roughly 90 million U.S. viewers use the toilet, sending about 1.4 billion liters of water into the sewage system in a short period, while on-site flushing can reach around 468,000 times per day. This brief but extreme peak stresses municipal water systems and, after purification and treatment, leaves a substantial "water footprint," adding pressure to regional water cycles and ecosystems.

In terms of solid waste, a regular NFL game produces about 40 tons of waste, whereas the Super Bowl generates nearly 60 tons [2]. The recycling/composting rate within this total is a key measure of waste management performance: a low rate implies that more recoverable materials are landfilled or incinerated, worsening resource loss and environmental pollution.

Greenhouse-gas emissions from the Super Bowl mainly come from transportation, spectator activities, and material promotion. Cross-regional travel by tens of thousands of spectators generates roughly 35,000 tons of CO_2e, while the advertising segment alone is estimated to emit about 2 million tons of CO_2—comparable to the annual emissions of around 1 million Americans. These figures show that greenhouse-gas emissions are a central component of the event's overall environmental footprint and have a significant cumulative impact on climate.

Table 2 Urban Location Characteristics Indicators

Category	Quantitative Indicator	Unit	Range
Climate	Average temperature in February	℃	[-5 ,20]
	Average precipitation in February	mm	[30,130]
Geography	density of population	Person/km^2	[50,4000]
	Average travel distance of the audience	Km/people	[200,2200]
Transportation infrastructure	Public transportation coverage rate	%	[15,70]
	The proportion of green transportation	%	[5,40]
Local energy structure	The proportion of renewable energy	%	[12,65]
	Electricity carbon intensity	kg CO_2e/kWh	0.25 ~ 0.55

Climate is a fundamental driver of event energy demand, so we use average February temperature and precipitation as core indicators. Geographical factors shape long-distance travel patterns, captured by population density and average spectator travel distance. A city's internal transport system determines the environmental efficiency of the "last mile," which we measure through public-transport coverage, share of green/low-carbon transport, and a traffic congestion index. Finally, the local energy structure sets the carbon footprint of all electricity use, reflected by the renewable-energy share and

grid carbon intensity. Together, these indicators highlight that the environmental footprint of a mega-event depends less on short-term organizational measures and more on the underlying characteristics of the host city.

4 PCA-Based Multiple Linear Regression Model of Environmental Impacts

Firstly, we use **Pearson correlations** to build a matrix including city features and environmental impact indicators to identify the linear relationships and potential multicollinearity. Secondly, we apply **principal component analysis** (**PCA**) on the standardized city features matrix, replacing highly correlated variables with **principal-component** scores. Lastly, we regress each environmental indicator on the selected component scores via **ordinary least squares** and evaluate fit using statistics such as $\boldsymbol{R^2}$, and **RMSE**. This approach provides a stable, interpretable mapping from "city features" to "environmental impacts" even with limited samples.

4.1 Data Collection and Processing

To further quantify the relationship between environmental impacts and city features, we use New Orleans (LIV) as a baseline, and compile four recent Super Bowl hosts, as shown in the Table 3.

Table 3 Recent Super Bowl host cities

Edition	Year	City	Venue
LIV	2020	Miami	Hard Rock Stadium
LIII	2019	Atlanta	Mercedes-Benz Stadium
LII	2018	Minneapolis	U.S. Bank Stadium
LI	2017	Houston	NRG Stadium

Based on host-city information listed above (let $\boldsymbol{n}$ denote the number of cities; here $\boldsymbol{n}$=4), we define the city feature variables (i.e. **Table 2**) as $X_1,\dots X_p, p=8$. Similarly, the environmental impact indicator (i.e. **Table 1**) is $Y_1,\dots Y_q, q=9$. For convenience in subsequent steps, we stack them into data matrices:

$$X=\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \in \mathbf{R}^{n\times p}, Y=\begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1q} \\ y_{21} & y_{22} & \cdots & y_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nq} \end{bmatrix} \in \mathbf{R}^{n\times q} \tag{1}$$

In the formulas, x_{ij} denotes the j-th city feature for city i, and y_{ij} denotes the j-th environmental impact indicator for city i. Since the indicators have different units and scales, we normalize each column of the data matrices.

$$z_{ij}=\frac{x_{ij}-\overline{x}_j}{s_j} \tag{2}$$

where z_{ij} is the normalized value, $\overline{x}_j$ the mean, and s_j the standard deviation. The same standardization is done for *Y*.

Because multicollinearity prevents accurate estimation of regression coefficients, we adopt the **correlation-matrix approach**, i.e., we examine the correlation matrix among variables to identify collinearity. For any two **normalized** variables (either a city-feature or an environmental-impact variable) V_i and V_j, the **Pearson correlation coefficient** $r_{ij}$is given by:

$$r_{ij} = \frac{Cov(V_i, V_j)}{s_i s_j} \tag{3}$$

where $Cov(V_i, V_j)$, is the covariance between V_i and V_j, and s_i and s_j are the standard deviations of the respective variables.

The purpose of correlation analysis is to preliminarily identify which city features are most closely linked to the environmental indicators and to detect whether serious multicollinearity exists among the city features, thereby providing a basis for subsequent **PCA**-based dimensionality reduction.

4.2 PCA-Based Multivariate Regression

To quantify how city features affect different categories of environmental impacts, we employ a PCA-based multivariate linear regression model. The specific steps are as follows:

Step1. Eigen-decompose the correlation matrix R:

$$R = P\Lambda P^T \tag{4}$$

where Λ is the diagonal matrix composed of eigenvalues λ_k (arranged in descending order), and P is the eigenvector matrix.

Step2. The k-th **principal component (PC)**, a linear combination of standardized features:

$$PC_k = p_{1k}Z_1 + p_{2k}Z_2 + \cdots + p_{pk}Z_p \tag{5}$$

where Z_j denotes the j-th normalized city-feature column vector.

Step3. For all cities, the **principal-component** score matrix is:

$$T = ZP \in \mathbf{R}^{n\times p} \tag{6}$$

where the k-th column, t_k, contains the scores of the k-th **principal component** across the four cities.

Step4. Compute the cumulative explained variance of the first K principal components. Here, we retain the **PCs** whose cumulative explained variance reaches at least 85%, and take these **PCs** as the new composite city-feature factors.

$$C_K = \frac{\sum_{k=1}^{K} \lambda_k}{\sum_{j=1}^{p} \lambda_j} \times 100\% \tag{7}$$

Step5. Next, we use the **principal-component** scores as the new explanatory variables and **perform principal component regression** (**PCR**) on the environmental impact indicators. Construct the score matrix of the first K principal components, augmented with an intercept.

$$X_{PC} = \begin{bmatrix} 1 & t_{1t} & \cdots & t_{1k} \\ 1 & t_{21} & \cdots & t_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & t_{n1} & \cdots & t_{nk} \end{bmatrix} \in \mathbf{R}^{n\times(K+1)} \tag{8}$$

Step6. For a single environmental indicator, the regression model is:

$$y_j = \beta_{j0}1 + \beta_{j1}t_1 + \cdots + \beta_{K1}t_K + \varepsilon_j \tag{9}$$

Step7. Fit the model by **ordinary least squares**:

$$\hat{\beta}_j = (X_{PC}^T X_{PC})^{-1} X_{PC}^T y_j \tag{10}$$

Step8. Finally, for each environmental indicator, compute the fitted values, the residual sum of squares, and the coefficient of determination R^2.

4.3 Results and Analysis

In this subsection, we use simulation results to quantitatively elucidate the relationship between environmental impacts and city features.

The correlation analysis between city features and environmental impacts is shown in Fig. 1. It reveals that host-city features, such as climate, transport systems, and energy mix exhibit strong correlations with event-period outcomes including energy use, water consumption, and carbon emissions. In particular, attendees' average travel distance is strongly positively correlated with audience- and transport-related emissions, whereas public-transit coverage and the share of green/low-carbon travel exert a suppressing effect on transport emissions.

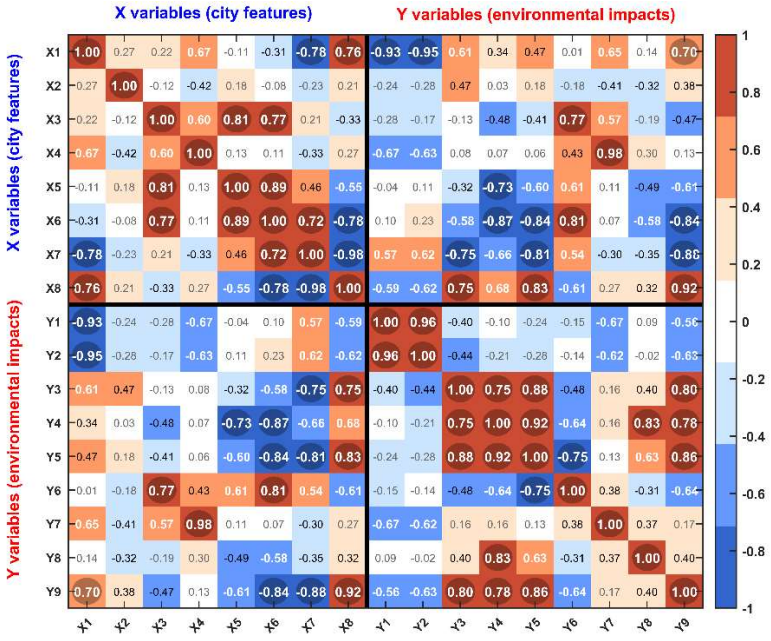


Figure 1 Heatmap of correlations between city features and environmental impacts

These patterns indicate that a host city's transportation conditions and spatial configuration are key determinants of the environmental footprint of mega-sport events like the Super Bowl.

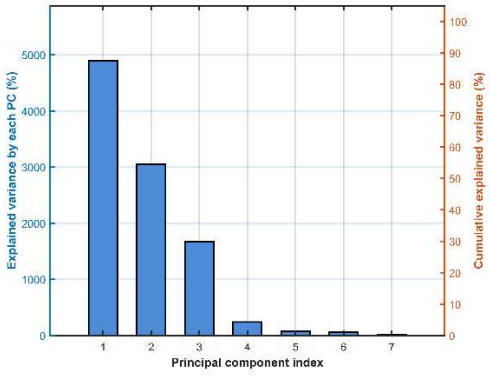


Figure 2 Principal components' explained-variance plot

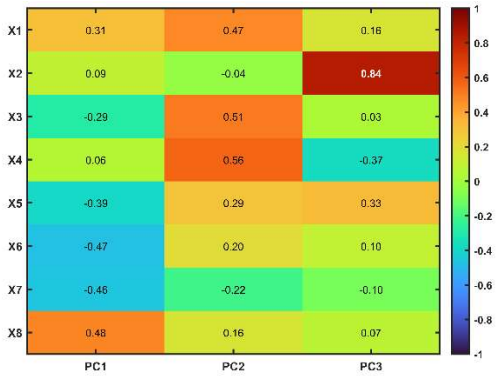


Figure 3 PCA loading heatmap

As shown in **Figure 2**, the first three principal components jointly explain over 85% of the total variance in the city-feature variables. Therefore, we retain the top three PCs as composite city-feature

indices for subsequent modeling. **Figure 3** displays the loadings of each feature on the PCs: PC1 is chiefly driven by population density, public-transit coverage, and the share of green/low-carbon travel, and can be interpreted as a "transport & urban intensity" factor. PC2 reflects climate conditions and grid carbon intensity, i.e., a "climate–energy structure" factor. PC3 has the highest loading on population density, confirming this as its core feature.

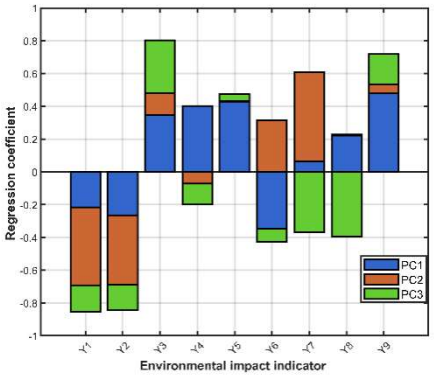


Figure 4 Decomposition of principal-component regression coefficients

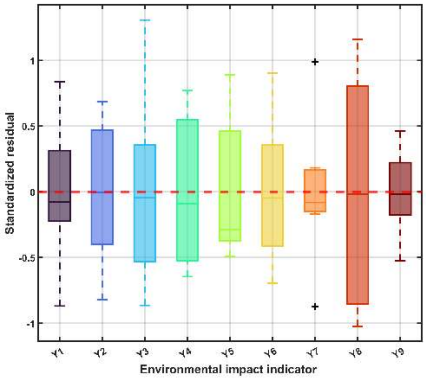


Figure 5 Standardized residual distributions for each environmental indicator

Figure 4 further shows that among the three latent dimensions extracted by PCA, the infrastructure & energy-structure component (PC1) primarily governs resource-use and waste-related indicators; the climate–distance component (PC2) dominates energy use and audience-related CO_2 emissions; and the mobility/density component (PC3) plays an important supplementary role in transport- and audience-related CO_2. The residual distributions in **Figure 5** are centered near zero with limited dispersion, indicating no systematic bias in the principal-component–based model and supporting its ability to reasonably characterize differences in environmental performance across host cities.

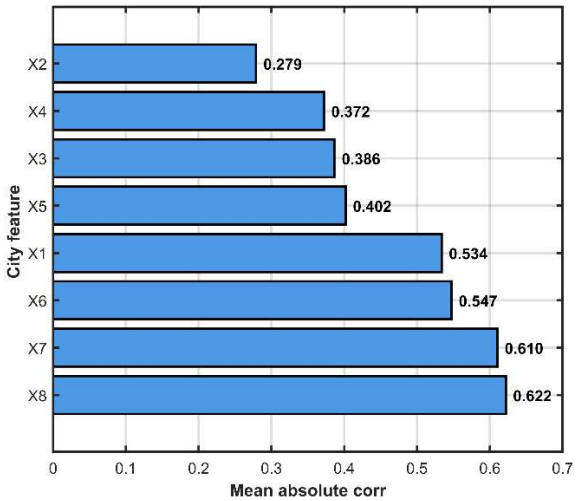


Figure 6 Importance ranking of environmental-impact drivers

When considering all environmental impact indicators $Y_1 - Y_9$, grid carbon intensity, the share of renewable energy, and the greenness of the transport system are the location traits with the most **system-wide** influence; next in importance are temperature and the level of public transit, while precipitation and other climatic details have comparatively weak average effects. Specifically:

Grid carbon intensity X_8 and renewables share X_9 rank highest in importance, indicating that with the same electricity use, materials, and event operations, outcomes differ systematically under

high-carbon vs. low-carbon grids. Share of green travel X_6 and public-transit coverage X_5 also rank near the top, showing that a city's transport mode mix jointly affects multiple environmental indicators (not only carbon emissions but also waste, recycling rates, water use, etc.).

Within climate variables, temperature X_1 matters more than precipitation X_2 : temperature is more tightly coupled to energy consumption and water use (HVAC loads, comfort needs), so its overall effect on the indicators is clear, whereas precipitation is not a core driver of cross-city differences.

Population density X_3 and average spectator travel distance X_4 have mid-to-low average effects, suggesting they are not the primary sources of variation in the environmental indicators.

This study employs **principal component analysis (PCA)** and **principal component regression (PCR)** to systematically quantify the **causal relationships** between city features and the environmental impacts of hosting mega-sport events. In the subsequent model-building stage, we will retain the indicator system outlined above as the baseline reference and use a quantitative index framework to characterize the events' environmental impacts. The indicators in this section also serve as references for later analyses of how differences in site features lead to variations in environmental impact.

5 Multi-Indicator Environmental Impact Assessment Model Based on PCR

In this section, we will establish a multi-indicator environmental impact assessment model based on the principal component regression model presented in Section 3. Notably, as COMAP requires determining which city should host the 2029 Super Bowl, the future environmental impact indicators cannot be known in practice. Therefore, we first use the ARIMA model to predict the characteristic indicators of different cities, then weight them with the trained regression coefficients from Section 3 to obtain the predicted environmental impact indicators. Finally, we apply the entropy weight method to weight the predicted indicators and derive the final comprehensive score. The process is as follows:

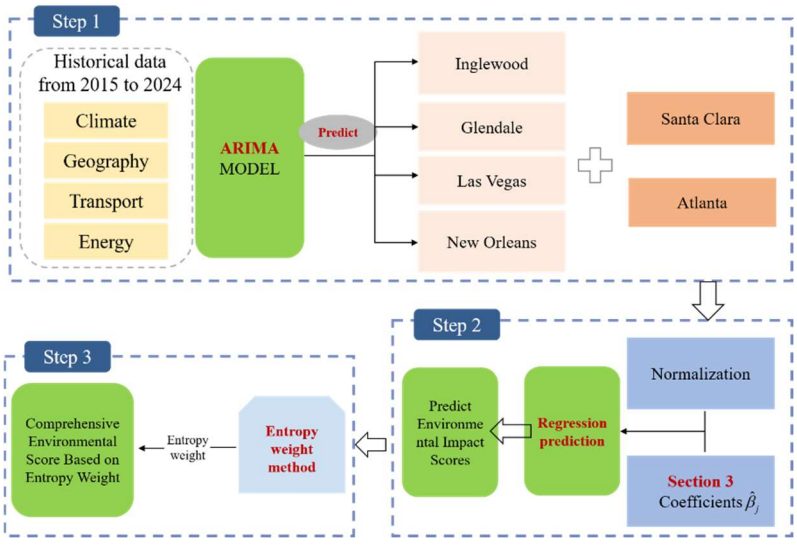


Figure 7 Multi-Indicator Environmental Impact Assessment Model Based on PCR

5.1 Data Collection and Prediction

Prior to the establishment of the model, it is crucial to clarify that the core of the problem lies in determining which city should host the 2029 Super Bowl. Regrettably, we do not have access to future data for direct analysis. Fortunately, predictions can be made based on historical data from previous

years. Therefore, we have collected the historical data regarding the local characteristic indicators of the six cities that have hosted the Super Bowl, which are listed in **Table 4**.

Table 4 Cities that have hosted the Super Bowl

Edition	Year	City
L	2016	Atlanta
LIII	2019	Santa Clara
LVI	2022	Inglewood
LVII	2023	Glendale
LVIII	2024	Las Vegas
LIX	2025	New Orleans

We have collected cities that currently have NFL teams but have never hosted the Super Bowl. There are 9 such cities in total, namely: Green Bay (home to the Green Bay Packers), Buffalo (home to the Buffalo Bills), Nashville (home to the Tennessee Titans), Cincinnati (home to the Cincinnati Bengals), Pittsburgh (home to the Pittsburgh Steelers), Cleveland (home to the Cleveland Browns), Kansas City (home to the Kansas City Chiefs), Baltimore (home to the Baltimore Ravens), and Charlotte (home to the Carolina Panthers). For the above - mentioned cities, we still need to select three potential host cities. Therefore, we have gathered the historical local characteristic data of these three potential cities, which are presented in **Table 5**.

Table 5 Potential Cities for Hosting the Super Bowl

Edition	Year	Potential host city
LXIII	2029	Santa Clara
LXIII	2029	Inglewood
LXIII	2029	Atlanta

Based on Assumptions 1 and 2, we can conclude that the historical climatic conditions of each candidate city during the Super Bowl period are relatively stable, and the urban population flow patterns remain consistent. Therefore, we only need to use the average values of temperature, precipitation, population density, and average spectator travel distance from 2015 to 2024 as proxies. For other indicators such as public transportation coverage, we will predict data for the next five years based on historical data from 2015 to 2024.

Next, we will adopt the ARIMA model to predict the future local characteristic data of candidate cities. For any city c and its local characteristic indicators m, we assume that their time series $y_{cm}(t)$ satisfy a specific model $ARIMA(p,d,q)$, which can be generally expressed as:

$$\Phi(B)\nabla^{d_{c,m}} y_{c,m}(t) = \mu_{c,m} + \theta(B)\varepsilon_{c,m}(t) \tag{11}$$

In the formula $\nabla^d y_t = (1-B)^d y_t$, the difference operator **B** is as follows: where is the lag operator, $\Phi(B)$ is the autoregressive polynomial, $\theta(B)$ is the moving average polynomial, $\varepsilon_{c,m}(t)$ is the white

noise error, and $d_{c,m}$ is the difference order, which is used to ensure that the differentiated sequence is stable.

For each sequence $\{y_{c,m}(t)\}_{t=1}^{T}$, under the given conditions $(p_{c,m}, d_{c,m}, q_{c,m})$, the parameter estimates are obtained through maximum likelihood estimation. Then, a least squares fit $t = 1,\ldots,T$ is used for fitting, and predictions are made $t = T+1,\ldots,T+H$. We need to predict up to the year 2029. Therefore H=5, there is:

$$\hat{y}_{c,m}(T+H) = a_{c,m} + b_{c,m}(T+H) \tag{12}$$

For the cities that have previously hosted events, as shown in Table 4. The predictions for the share of renewable energy, power carbon intensity, public transportation coverage rate, and green transportation share were made using ARIMA (1,1,0), ARIMA (0,1,0), ARIMA (0,2,0), and ARIMA (0,2,1) respectively. The results are as follows:

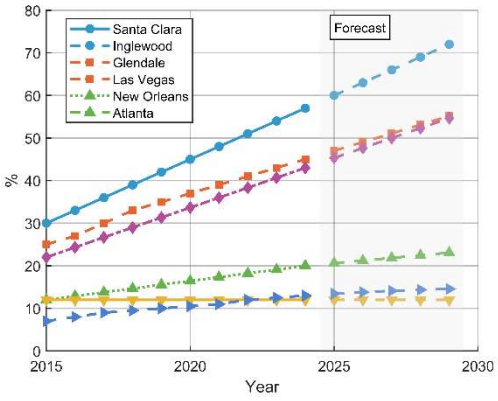


Figure 8 Prediction of Renewable Energy Share

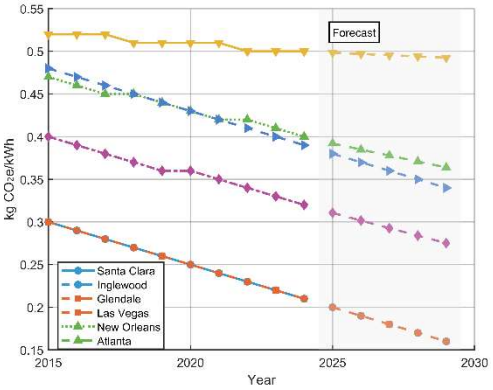


Figure 9 Prediction of Power Carbon Intensity

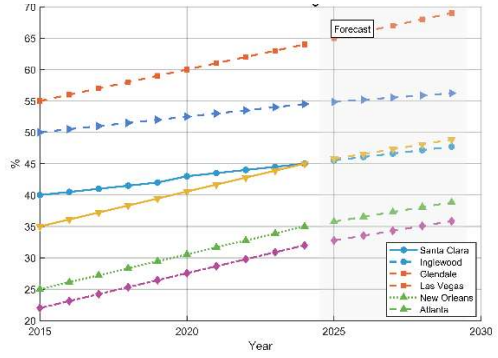


Figure 10 Prediction of Public Transportation Coverage Rate

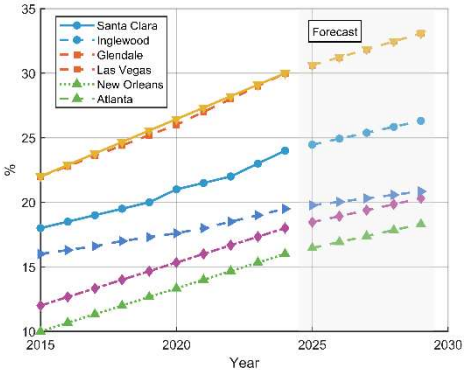


Figure 11 Prediction of Green Transportation Share

Similarly, for cities that have never held such events, namely those in Table 5, we also used the ARIMA model for prediction. The results are as follows:

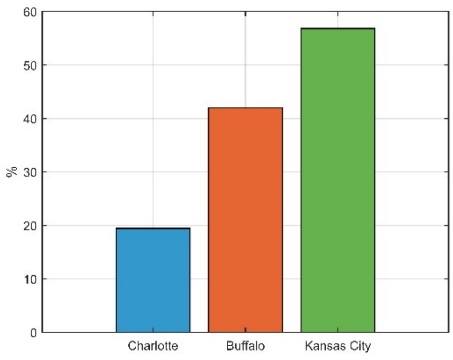


Figure 12 Proportion of Renewable Energy in 2029 Forecast

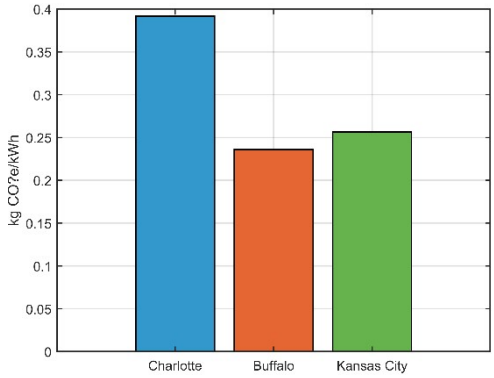


Figure 13 Prediction of Electricity Carbon Intensity in 2029

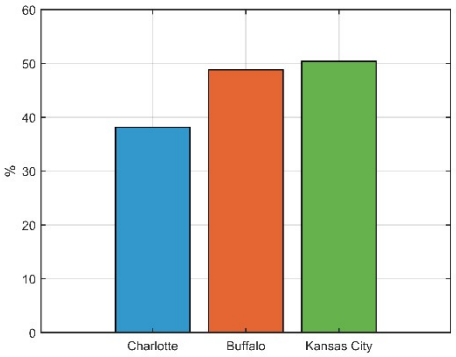


Figure 14 Prediction of Public Transportation Coverage in 2029

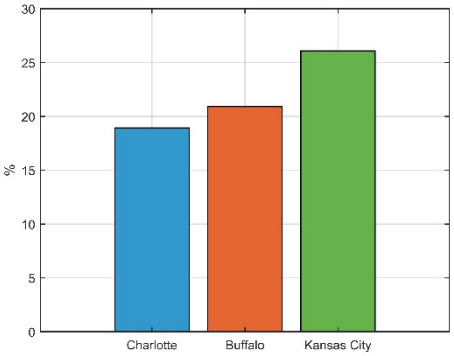


Figure 15 Prediction of Green Transportation Share in 2029

5.2 Overall Environmental Impact Score

In this subsection, we employ the entropy weight method to determine the weights of environmental impact indicators, as specified in **Table 1**. Notably, since the environmental impact indicators for the upcoming event are unavailable, we utilize the principal component regression model established in **Section 3**—substituting the predicted urban characteristic data from Subsection 4.1 into this model to generate predicted values $\hat{y}_j$, which are defined as "environmental scores". Finally, weighting is performed using the **entropy weight method**.

Prior to implementing the entropy weight method, it is necessary to clarify the positive and negative attributes of indicator Y: a higher value of a positive indicator signifies a smaller environmental impact, while a higher value of a negative indicator indicates a larger environmental impact. The specific calculation steps for the comprehensive environmental impact score are outlined below:

Step 1. Normalize the positive and negative indicators separately, substitute the predicted urban characteristic data X into the model for prediction, and obtain $\hat{Z}_Y = (\hat{z}_{ik})_{m\times n_Y} \in [0,1]$ the normalized matrix.

Step 2. Normalize each column of indicators into a "probability distribution" to characterize the relative magnitude of each city's performance on that indicator, and define a probability matrix for each environmental indicator k.

$$p_{ik} = \frac{\hat{z}_{ik}}{\sum_{i=1}^{m} \hat{z}_{ik}}, i = 1, \cdots, m \tag{13}$$

Step 3. For the k-th column of the environmental impact indicators, the information entropy is defined as:

$$e_k = -\frac{1}{\ln\ m}\sum_{i=1}^{m} p_{ik} \ln\ p_{ik}, 0 \le e_k \le 1 \tag{14}$$

Step 4. The redundancy d_k is defined as follows: a larger value d_k indicates greater variability in indicator k and a stronger contribution to distinguishing between cities.

$$d_k = 1 - e_k \tag{15}$$

Step 5. The entropy weight of the environmental impact indicator is defined as:

$$w_k^Y = \frac{d_k}{\sum_{k=1}^{n_Y} d_k}, \sum_{k=1}^{n_Y} w_k^Y = 1 \tag{16}$$

Step 6. For each city i, based on the entropy weights, calculate the weighted sum of its scores $\hat{z}_{ik}$ across all environmental impact indicators to obtain the comprehensive entropy-weighted environmental score.

$$\hat{F}_i^Y = \sum_{k=1}^{n_Y} w_k^Y \hat{z}_{ik} \tag{17}$$

Step 7. Finally, normalizing the comprehensive scores yields:

$$S_i = \frac{\hat{F}_i^Y - \min_r \hat{F}_r^Y}{\max_r \hat{F}_r^Y - \min_r \hat{F}_r^Y}, i = 1, \ldots, m \tag{18}$$

5.3 Model Results and Analysis

5.3.1 Has previously hosted the city's scoring results

First, we conduct entropy weight analysis on the 9 environmental impact indicators of the 6 candidate cities (Santa Clara, Inglewood, Las Vegas, Glendale, Atlanta, and New Orleans), yielding the weight distribution shown in Figure 16. The results indicate that indicators such as venue water consumption, per capita water consumption, spectator carbon emissions, and transportation carbon emissions have the highest weights, followed by electricity consumption indicators related to venue and hotel energy use. This suggests that among the six cities, the differences in water resource consumption and spectator/transportation-related carbon emissions are the most significant, serving as key factors distinguishing the environmental performance of the cities. In contrast, indicators with smaller intercity differences (e.g., material carbon emissions) have lower entropy weights, exerting limited impact on the comprehensive score and primarily functioning as supplementary constraints. This is highly consistent with the actual situation of large-scale sports events—water resource management, spectator travel, and transportation organization should be the core links of environmental control.

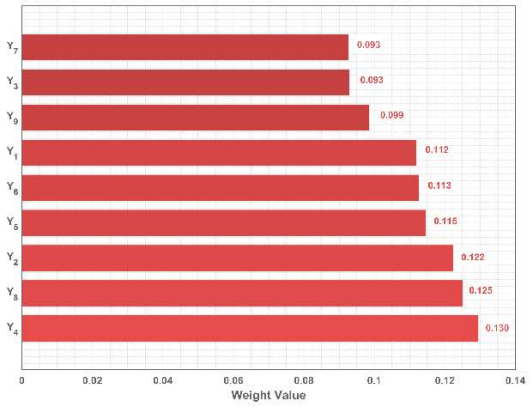


Figure 16 Distribution of Weighting Factors for Environmental Impact Indicators

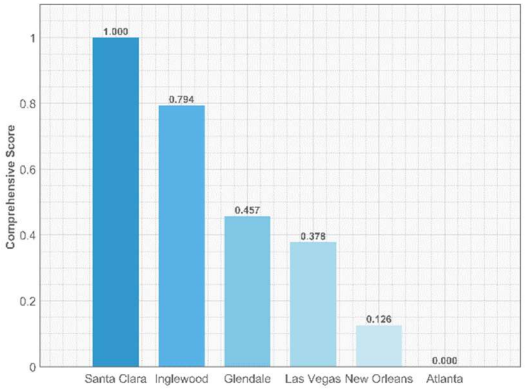


Figure 17 Final City Score

Building on this, we calculated the weighted sum of the 9 environmental indicators using the entropy weight method to obtain the comprehensive environmental score for each city, with the ranking results presented in **Figure 17**. Overall, Santa Clara demonstrates the best environmental performance, excelling in renewable energy share, power carbon intensity, water use control, waste management, and recycling rates. It holds a particularly notable advantage in spectator and transportation carbon emissions, thus achieving the highest score when environmental footprint is the sole evaluation criterion and emerging as the most suitable city to host the Super Bowl from an environmental perspective. Inglewood follows closely behind: leveraging California's relatively clean power structure and well-developed public transportation system, its overall environmental performance is only slightly inferior to Santa Clara, making it a high-quality alternative venue. Las Vegas and Glendale fall into the middle tier; while they have strengths in certain dimensions, their overall environmental load is moderately higher due to elevated energy and transportation demands. Atlanta and New Orleans, however, face disadvantages in most key negative indicators—characterized by high power carbon intensity, insufficient renewable energy share, and relatively high levels of water consumption and carbon emissions. Consequently, their comprehensive environmental scores are relatively low, and they are not recommended as priority host cities from a strict environmental sustainability perspective.

5.3.2 The result of city that has never hosted the event

After conducting entropy weight analysis on the 9 environmental impact indicators of three cities with NFL teams but no prior Super Bowl hosting experience—Charlotte, Buffalo, and Kansas City—**Figure 18** shows that the entropy weight coefficients of indicators such as venue water consumption, per capita water consumption, spectator carbon emissions, and transportation carbon emissions are significantly higher than those of other indicators. This is followed by indicators including venue/hotel electricity consumption and total waste generation, while indicators with relatively small differences among the three cities (e.g., material carbon emissions) have lower weights and exert limited marginal impact on the final score.

Building on the entropy weight results in **Figure 18**, we calculated the weighted sum of the 9 environmental impact indicators to obtain the comprehensive environmental score for each of the three cities, with the ranking presented in **Figure 19**. The results indicate that Kansas City achieves the highest comprehensive environmental score, making it the optimal potential host city among the three

candidates.

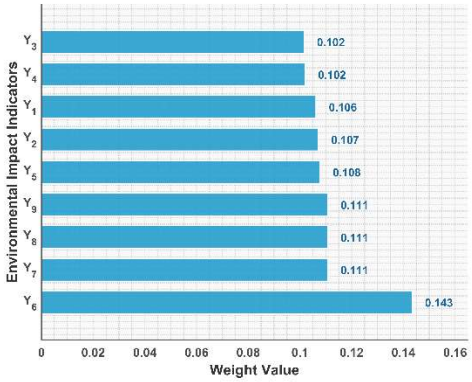


Figure 18 Distribution of Weighting Factors for Environmental Impact Indicators

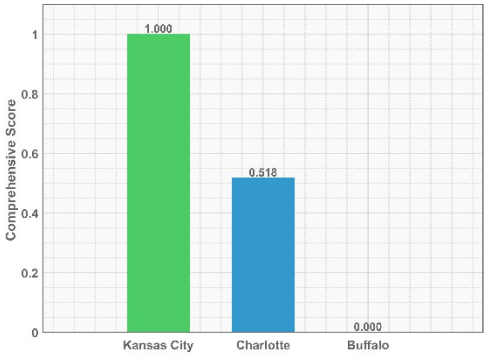


Figure 19 Final City Score

6 Comprehensive Venue Evaluation Model Based on PCR

In this subsection, we extend the previously developed **principal-component regression (PCR) multi-indicator model** for evaluating environmental impacts. On this basis, we incorporate the intrinsic characteristics of multi-purpose venues and sustainability-oriented impact metrics, thereby constructing a **PCR**-based comprehensive venue evaluation model. We also identify the dominant environmental factor that most affects the sustainability of impacts for a single mega-sport event compared with a multi-sport comprehensive event.

6.1 Evaluation Criteria and Analysis

Before building the model, we clarify the core question: we need to extend our earlier framework to cover other types of sporting events. Here, we consider the **Summer Olympic Games** as a comprehensive multi-sport event. The Summer Olympics are held every four years in July–August, with a competition period of about 17 days. On top of the previous environmental-impact indicators and site characteristics, we add several indicators to capture features specific to comprehensive events, as shown in **Tables 6 and 7**.

Table 6. Additional environmental-impact indicators for the Summer Olympics

Category	Quantitative indicator	Unit	Range
Energy consumption	**Olympic Village electricity intensity**	MWh	$3\times10^3 \sim 2\times10^4$
Waste generation	**Construction & demolition waste from facilities**	t	$5\times10^4 \sim 3\times10^5$ t

In comprehensive events, the **Olympic Village** operates continuously for weeks and integrates accommodation, dining, medical services, training, and offices—essentially a "small city." Introducing the indicator of **Olympic Village electricity intensity** captures one of the key environmental challenges that distinguishes week-long, multi-functional events from single-day events.

For events like the Olympics that require large-scale new construction or renovation of venues, the Village, and supporting infrastructure, **construction and demolition (C&D) waste** forms a non-

negligible part of the life-cycle environmental impact. By quantifying "how much was built for the event" and "how much was torn down," we incorporate resource inputs and waste into the evaluation. This enables the model to assess not only operational-phase pressures but also the **full life cycle** of infrastructure—from groundbreaking to decommissioning—which better reflects the sustainability factors of comprehensive multi-sport events.

Table 7. Newly added site-characteristic indicators for host cities

Category	Quantitative indicator	Unit	Range
Climate	Average temperature **(July-August)**	℃	8 ~ 28 ℃
	Average precipitation **(July-August)**	mm	0 ~ 200 mm
Transport & Infra-structure	**International airport throughput**	million passengers	1500 ~ 12000
Venue construc-tion	**Mean inter-venue travel distance**	Km	5 ~ 30 km
	In-Games venue reuse rate	%	40% ~ 80%
	Post-Games venue reu-tilization rate	%	50% ~ 95%

"International airport throughput" reflects a city's node rank and carrying capacity within the global air network and thus has structural significance for the environmental impacts of comprehensive multi-sport events. Including it as a site characteristic helps capture system-level global accessibility—an environmental factor rarely considered for single-sport events that are largely fed by domestic flows.

Compared with single-venue events, a defining feature of comprehensive games is the high frequency of movements by athletes, officials, media, and spectators among multiple venues. Therefore, the mean inter-venue travel distance becomes a key structural variable affecting transport energy use and carbon emissions.

The in-Games venue reuse rate measures, within a single edition, the extent to which the same venues are used for different events or sessions—an important indicator of scheduling efficiency and spectator routing, and a practical metric for sustainability planning during the Games.

In contrast, the post-Games venue reutilization rate directly concerns the entire life cycle of venues. Combined with long-term population and site traits, it allows analysis or proxy estimation of whether and how long facilities will continue to be used after the Games. A higher reutilization rate implies lower hidden costs and a stronger legacy, turning a short-term event into an ongoing environmental stewardship responsibility for the host city.

6.2 Results of the Comprehensive-games Evaluation Model

This model follows the same framework as the previous section, but **adds** several environmental-impact indicators and city/site characteristics to capture the distinct features of comprehensive multi-sport events. We therefore present results and analysis directly.

Figure 20 reports the loadings of site characteristics $X_1 - X_{12}$ on the first three principal components (PC1–PC3). PC1 loads heavily on public-transit coverage (X_5), share of green travel (X_6), renewables share (X_9), and grid carbon intensity (X_9), and can be interpreted as an **"**infrastructure–transport–energy-sustainability**"** dimension. PC2 reflects climate and demographic structure—e.g., temperature (X_1), precipitation (X_2), and population density (X_3)—i.e., a **"**climate/locational heterogeneity**"** dimension. PC3 has large loadings on mean inter-venue distance (X_{10}) and in-Games/post-Games venue reuse rates (X_{11}, X_{12}), and is thus a **"**venue-network layout & resource reutilization**"** dimension. This structure aligns with the three major drivers of environmental impact in Olympics-type comprehensive events: (1) the city's overall low-carbon infrastructure, (2) local climate–energy conditions that shape heating/cooling needs, and (3) the spatial organization of the venue network and the reuse potential of the Olympic Village.

From **Figure 21**, the indicators with the largest weights include operational energy use of venues and the Olympic Village (Y_1, Y_3), construction & demolition (C&D) waste (Y_8), and audience (Y_9)-transport-related CO_2 emissions (Y_{10}). These map directly to the distinct sustainability challenges of comprehensive games: (i**) multi-venue, weeks-long energy and water demand centered on the Village**; (ii) **materials use and embodied/carbon emissions from new builds, retrofits, and post-Games modifications**; and (iii) **multi-tier transport systems for international spectators and inter-venue commuting**. **Entropy-weighting** results indicate that, under a purely objective weighting scheme, the model naturally assigns greater weight to long-term, structural pressures rather than short, minor components, which directly aligns with our understanding of the environmental characteristics of comprehensive events such as the **Summer Olympics**.

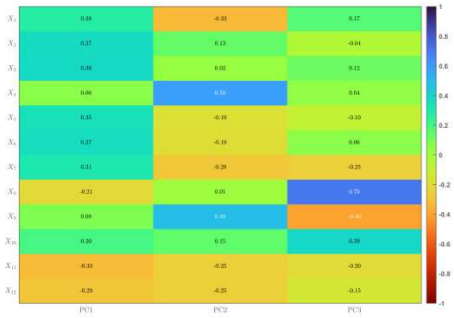


Figure 20 PCA loading heatmap

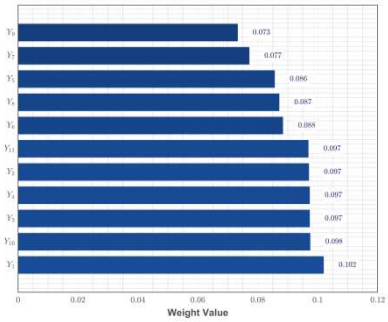


Figure 21 Weights of environmental-impact indicators

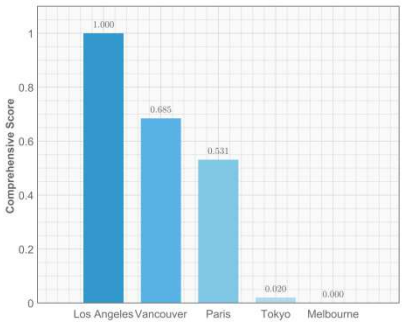


Figure 22 Final city scores

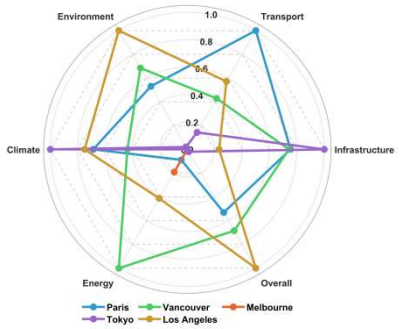


Figure 23 Radar chart of multi-dimensional evaluation

Figure 22 presents the bar chart of the final rankings after normalizing the composite environmental-performance scores. As shown, **Los Angeles** attains the highest overall score, with a clear lead over the second-ranked city, while the remaining cities follow in descending order. **Figure 23** uses a radar chart to display the relative strengths and weaknesses of the five cities across six dimensions: infrastructure, transport, environment, climate, energy, and overall performance.

7 Comparative Analysis of Model Results

By comparing the results from Sections 3–4 with those from Section 5, several structural differences emerge between single-event mega-events such as the Super Bowl and large-scale, multi-venue or multi-sport competitions such as the Summer Olympic Games. The key differentiating mechanisms can be summarized as follows:

(1) Temporal Structure: Short-Term Peaks vs. Long-Horizon Accumulation

In the Super Bowl model (**Sections 3–4**), the entropy-weighted results assign the highest weights to stadium water use, per-capita water consumption, spectator travel emissions, and transportation-related emissions, followed by electricity consumption for stadiums and hotels. Construction-phase and long-term legacy indicators contribute almost nothing to the weighting structure. This pattern indicates that the environmental pressure of a single-event competition is dominated by short-duration, peak-load consumption concentrated around game day. In contrast, in the Olympic model, indicators such as Olympic Village electricity intensity and construction/demolition waste obtain significantly higher entropy weights. This shift reflects that multi-venue events generate cumulative, time-extended sustainability pressure, where long-horizon resource use becomes more influential than short-term operational spikes.

(2) Spatial Structure: Single-Venue Events vs. Multi-Venue Networks

In the Super Bowl model, spatially relevant factors are primarily: **Average spectator travel distance, Public transport coverage, Share of green transportation.** Regression results show that these variables strongly explain spectator and transportation-related carbon emissions. In the Olympic model, newly added variables—average inter-venue travel distance, in-Games venue reuse rate, and post-Games reutilization rate—form a distinct principal component (PC3), interpreted as the "venue spatial configuration and legacy utilization" dimension. This component contributes meaningfully to multiple environmental indicators. This comparison highlights that for comprehensive multi-sport events, the spatial network of venues, together with the multi-tier mobility system linking venues and the Olympic Village, becomes a structural driver of environmental outcomes.

(3) Shared Core Drivers: Electricity Systems and Transport Structure

Across both models, a consistent result emerges: **Electricity carbon intensity, Renewable-energy share, Public transport coverage, and Green mobility proportion.**All exhibit high importance in both the principal component regressions and the entropy weighting scheme. These variables function as common core determinants of environmental performance. However, the mechanisms differ. For the Super Bowl, these city attributes primarily amplify or mitigate single-day carbon footprints via short-term electricity demand and transportation organization. For the Olympic Games, they determine not only operational emissions during the 17-day event window but also the long-term emission baseline for decades of venue and Olympic Village operation.

8 Sensitivity Analysis

In this subsection, to characterize how infrastructure investment, energy procurement, and changes in transportation modes affect the performance of cities bidding to host large-scale sports events in the future, we build on the original model in Section 5. Specifically, we treat infrastructure investment, energy procurement, and transportation modes as adjustable policy variables, mapping them to parameterized changes in the urban characteristic matrix X and the environmental indicator matrix Y. Subsequently, we recalculate the comprehensive environmental scores under the same entropy weight system and compare the score differences before and after the scenario simulation to quantitatively evaluate the marginal improvement effects of different policy combinations on urban sustainability and bidding competitiveness.

We consider a total of c candidate cities, with the original baseline urban characteristic vector denoted as $X_c^{base} = (X_{c1}^{base}, \ldots, X_{c12}^{base})$ and the baseline environmental impact indicator vector as $Y_c^{base} = (Y_{c1}^{base}, \ldots, Y_{c11}^{base})$. We aim to modify the urban characteristics and environmental indicators through three types of strategy variables: infrastructure investment intensity $u_c^{(I)} \in [0,1]$, green energy procurement intensity $u_c^{(E)} \in [0,1]$, and transportation mode transformation intensity $u_c^{(T)} \in [0,1]$. The urban characteristics and environmental indicators under the aforementioned policies are denoted as, respectively.

$$X_c^{pol} = f_X(X_c^{base}, u_c^{(I)}, u_c^{(E)}, u_c^{(T)}) \tag{19}$$

$$Y_c^{pol} = f_Y(Y_c^{base}, X_c^{pol}, u_c^{(I)}, u_c^{(E)}, u_c^{(T)}) \tag{20}$$

where all functions denote linear functions $f_x(\bullet), f_Y(\bullet)$—specifically, the strategy variables linearly increase or decrease X_c^{base} and Y_c^{base}.

We denote the score under the baseline environmental impact indicators as $S_c^{base} = F(Y_c^{base})$, and the score after introducing the policies as $S_c^{pol} = F(Y_c^{pol}(u_c^{(I)}, u_c^{(E)}, u_c^{(T)}))$. Therefore, the policy-driven improvement in the city's score can be expressed as: $\Delta S_c = S_c^{pol} - S_c^{base}$.

Next, we discuss the impacts of the three types of strategies.

(1) Infrastructure Investment Intensity $u_c^{(I)}$

- Strategies: Venue energy-saving renovation, smart water management, waste recycling facilities, and public transportation expansion.
- Impacts on urban characteristics X: Increasing public transportation coverage rate X_5, proportion of green transportationX_6, venue reuse rate X_{11}, and post-event utilization rate X_{12}; reducing the average travel distance between venues X_{10}.
- Impacts on environmental indicators Y: Reducing venue electricity consumptionY_1, Olympic Village electricity consumption intensity Y_3, venue-level water consumptionY_4, and waste from facility construction and demolition Y_8.

(2) Green Energy Procurement Ratio $u_c^{(E)}$

- Strategies: Purchasing renewable electricity, increasing investment in local photovoltaic/wind power, etc.
- Impacts on urban characteristics X: Increasing the proportion of renewable energy in electricityX8 and reducing power carbon intensity X9.
- Impacts on environmental indicators Y: Reducing venue electricity consumptionY1, hotel electricity consumption Y2, and Olympic Village electricity consumption Y3.

(3) Transportation Mode Transformation Intensity $u_c^{(T)}$

- Strategies: Rail transit connection, temporary public buses, car-free day policies, etc.

- Impacts on urban characteristics X: Increasing public transportation coverage rateX5， and proportion of green transportation X6.
- Impacts on environmental indicators Y: Reducing transportation carbon emissions Y10.

We reflect the aforementioned impacts through linear weighting to quantify the effects of the strategies. We design three sets of distinct strategies to assess how changes in infrastructure investment, energy procurement, and transportation modes influence cities bidding to host large-scale sports events in the future. Different combinations of strategy investment intensities are established, as shown in Table 8. The simulation results under different strategies are presented as follows:

Table 8: Strategy Intensity Corresponding to Candidate Cities

City	The first group of policy intensity	The second group of policy intensities	The third group of policy intensity
Paris	[0.4, 0.3, 0.3]	[0.2, 0.3, 0.3]	[0.7, 0.7, 0.6]
Vancouver	[0.5, 0.6, 0.4]	[0.2, 0.3, 0.3]	[0.2, 0.3, 0.3]
Melbourne	[0.6, 0.4, 0.3]	[0.2, 0.3, 0.3]	[0.2, 0.3, 0.3]
Tokyo	[0.5, 0.3, 0.5]	[0.2, 0.3, 0.3]	[0.2, 0.3, 0.3]
Angeles	[0.7, 0.7, 0.6]	[0.7, 0.7, 0.6]	[0.2, 0.3, 0.3]

By adjusting the intensity of the strategies, we find that increasing the investment intensity of the corresponding strategies for a city leads to an increase in its respective urban score, which aligns with our expected outcomes. In the figure below, red indicates a negative impact $\Delta S_c = S_c^{pol} - S_c^{base}$ after strategy adjustment, while green indicates that the strategies have exerted a positive effect.

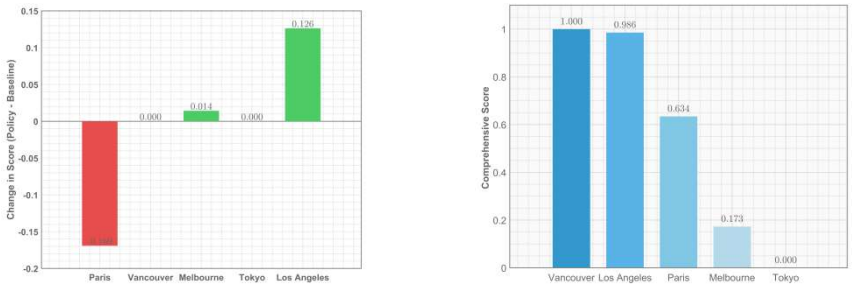


Figure 24 The increase in city scores and ranking under the first group of policy-driven measures

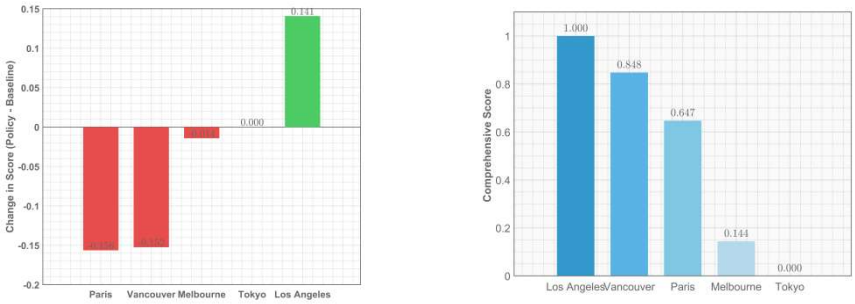


Figure 25 The increase in city scores and ranking under the second group of policy-driven measures

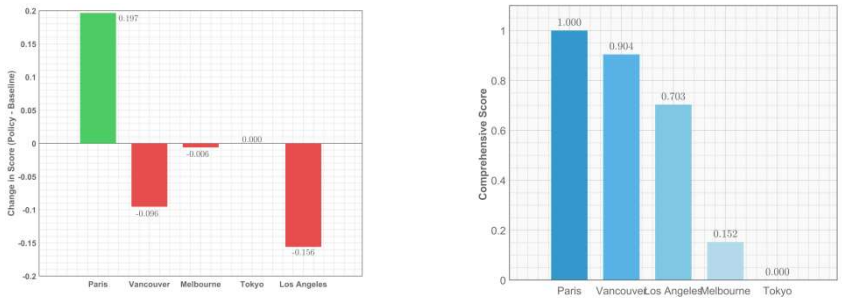


Figure 26 The increase in city scores and rankings under the third group of policy-driven measures

9 Strength and Limitations

The merit of our model lies in its scientifically grounded indicator system and the integration of multiple analytical methods. By combining indicators for energy use, water, waste, transport emissions, and the urban environment, it offers a multi-dimensional yet intuitive way to evaluate the Super Bowl's environmental impact. The PCA-based regression model mitigates multicollinearity and extracts three stable principal components explaining over 85% of total variance, giving both statistical robustness and clear interpretation. ARIMA forecasting combined with PCA regression allows forward-looking evaluations of candidate host cities without requiring future observations, while the entropy-weight method enhances objectivity by assigning greater weight to more discriminative indicators. Scenario-based sensitivity analysis further shows how policy changes—such as expanded renewable energy or shifts in transport structure—alter projected impacts, making the framework useful for long-term sustainability planning.

However, the framework has limitations. The regression is calibrated on a relatively small set of historical host cities, which constrains statistical power and generalizability. Several indicators rely on simplified assumptions, such as multi-year climate averages and linear mobility approximations, which may bias results for cities with high variability or rapid structural change. ARIMA forecasting assumes stable trends and does not capture non-linear shocks from policy shifts, infrastructure upgrades, economic change, or extreme climate events, potentially distorting projections for 2029. The assessment also focuses mainly on operational impacts, only partially accounting for embodied carbon in stadium construction and long-term energy use from subsequent events, so the full life-cycle footprint is understated. Finally, while entropy weighting is objective, it ignores expert judgment and may underemphasize important but low-variance indicators; future work that blends objective and expert-based weights could improve realism and balance.

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A Letter to the NFL

Dear Members of the NFL Super Bowl Selection Committee,

On behalf of the HiMCM Super Bowl Sustainability Consultants (SSC), we are honored to present our recommendation for the host city of the 2029 Super Bowl. Based on a quantitative analysis focused on environmental impact, our team evaluated how different cities would perform if they hosted the game in 2029.

In our study, we compared candidate cities on key dimensions such as event-related energy use, greenhouse-gas emissions, water consumption, and solid-waste generation, together with city factors including climate, electricity mix, transportation options, and existing sustainability practices. Using these indicators, we estimated the overall environmental footprint of a Super Bowl in each city and ranked them accordingly.

From this comparison, we respectfully recommend **Santa Clara, California** as the 2029 host city. Santa Clara stands out for three main reasons:

- Lower-carbon energy and mild climate**:** A relatively clean regional grid and temperate February weather reduce energy demand for the stadium, hotels, and other facilities, cutting associated emissions.
- Efficient transportation structure**:** As part of a dense, transit-served metropolitan area with multiple airports, Santa Clara allows more spectators to rely on shorter trips and public transport rather than long-haul travel and private cars.
- Strong water and waste management: Established conservation measures, reclaimed-water use, and advanced recycling and composting systems make ambitious water-saving and high waste-diversion targets realistic during the event week.

Other cities such as Las Vegas, New Orleans, and Glendale have strong football traditions and valuable hosting experience. However, under consistent environmental criteria, our results indicate that Santa Clara generates a smaller projected footprint while still meeting the operational needs of a modern Super Bowl.

Selecting Santa Clara for the 2029 Super Bowl would help the NFL significantly reduce the event's environmental impact and demonstrate clear leadership in aligning elite sport with sustainability goals.

Thank you for your time and consideration.

Yours sincerely,
The HiMCM Super Bowl Sustainability Consultants (SSC) Team

AI Use Report

ChatGPT 5.1 Model was used solely to polish the text. No generative AI tools were employed in the creation of core content, methodology, and programming.